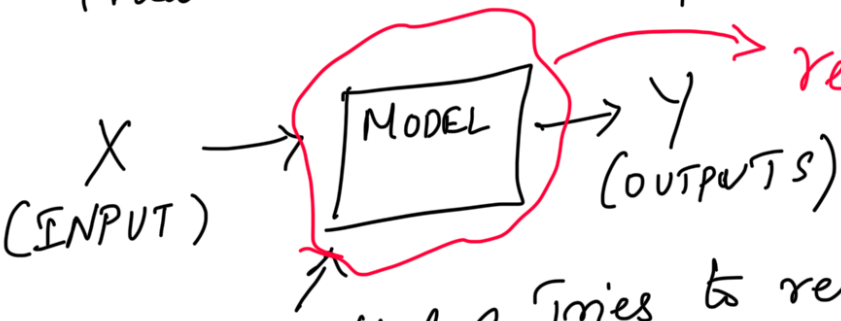
AGENDA:

Traditional ML → Deep learning → ChatGPT (GenAI)



Mathematical mapping $y = f(x)$ tries to remember/Memorising Pattern in my data.

① Pattern Recognition ② Representation Learning
② Information Retrieval ④ Machine learning
(Info. Systems)

---

① Python. | Tool with which we code these algos. ✓ (language)

② Stats & Probability | ⇒ foundations of these algorithms.

③ Artificial Intelligence




⇒ Sort → If else Statements | If colour red & shape roundish (Rule) → Yes → Apple ; No → Banana.

↳ i) Rule-Based Systems.

↙ Possibility of patterns becomes hard to describe!

GMAIL – SPAM.

SPAM

① 'FREE' AIRTEL "Congrats! You have won free iphone"



Airtel - Subscribe

→ IF | ELSE

INCEPTEZ
↳ Join free Session
Tomorrow 9.

How many rules should I write?
10? 100? 10,000? → RULE EXPLOSION Problem!!

NOT SPAM

Inferential Statistics.
↳ Rule → Give Output
(Predefined)

Machine Learning Systems
↳ Examples → Learn patterns → Learned Model

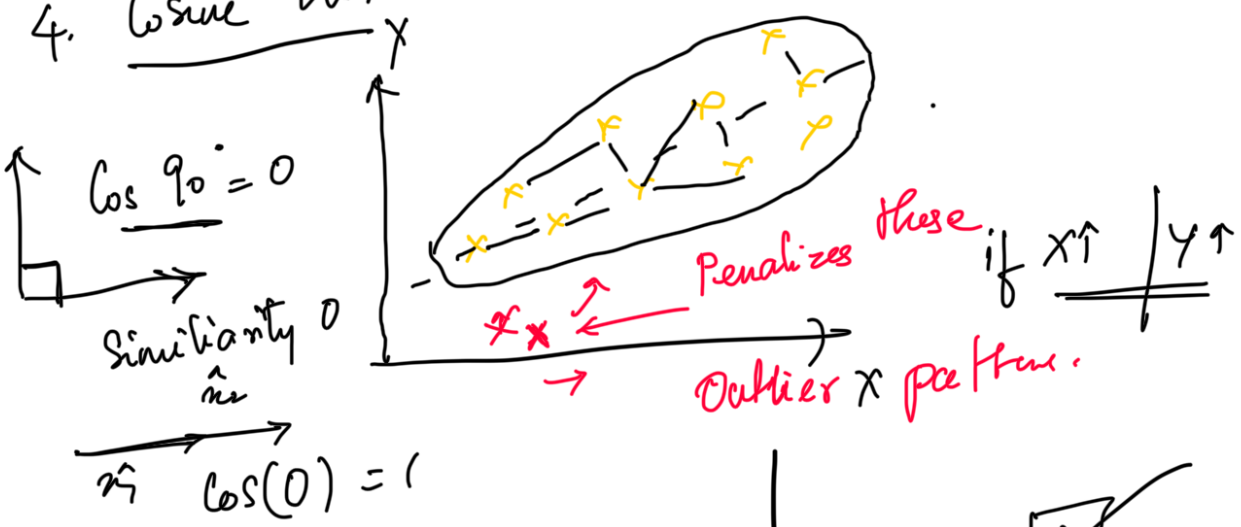
DATA
↓ Preprocessing
HUMAN-ENGINEERED features. (Correlations)
(feature Engineering)
↓
Machine Learning Models.
↓
Classification / Regression.

Traditional ML → learns predictive relationships from features. often, "designed & selected" by humans

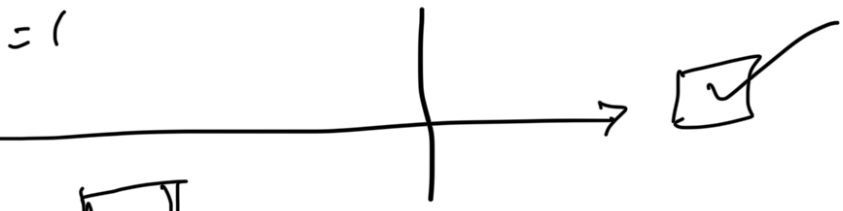
Regression
① linear Regression (done)
② Polynomial Regression (done) → Poly-logistic?
③ KNN → KNN

Classification
Logistic Regression

1. Euclediean Dist → $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
2. Manhattan Dist → $|x_2-x_1|+|y_2-y_1|$ → L1
3. Mahalanobis Dist → $x-\mu\ \Sigma^{-1}(x-\mu)^T$
4. Cosine dist

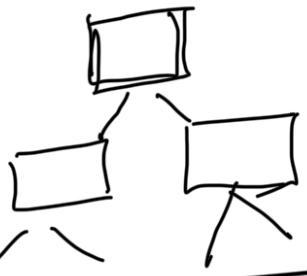


(4) Decision Tree



Ensemble

(5) R.F ← BAGGING



BOOSTING

(6) AdaBoost — NOT IN OUR SYLLABUS

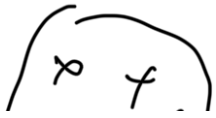
(7) GBM — Good to know

(8) XGBoost — Good to know.

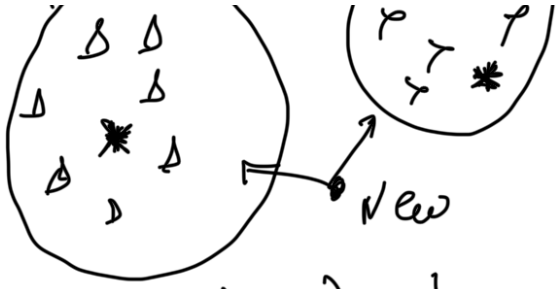
(9) SVM — Adv. ML - Good to know

(10) Naive Bayes | NOT NEEDED

UNSUPERVISED LEARNING



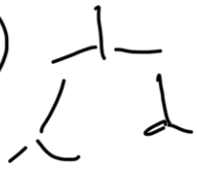
① K-Means Clustering Algo



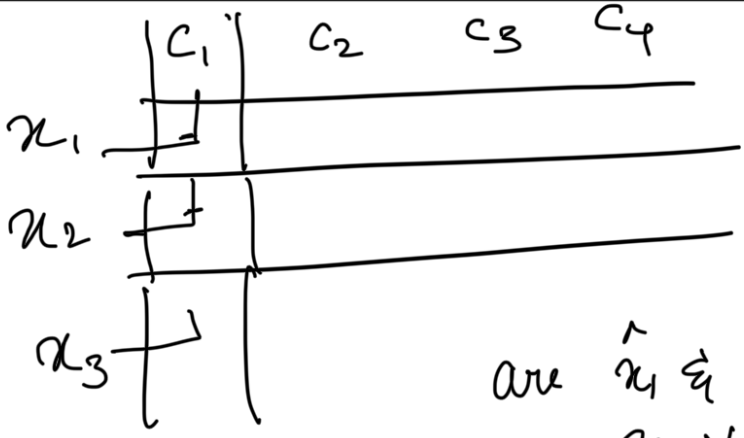
② HIERARCHICAL CLUSTERING

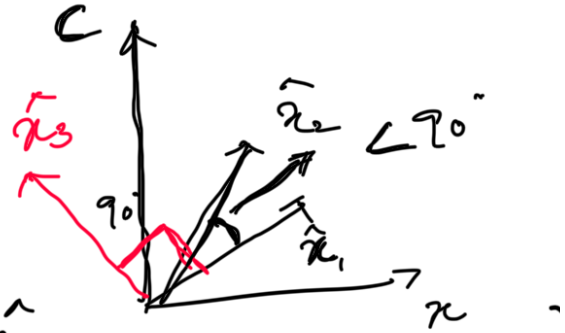
↳ Agglomerative (Good to know)

↳ DIANA / Conglomerative

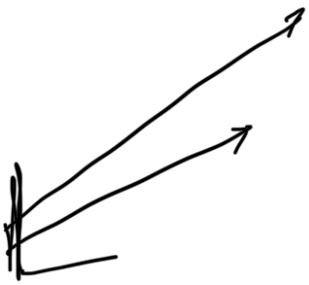


③ DBSCAN ⟶ (Good to know).





are $\hat{x}_1$ & $\hat{x}_3$ Similar? No! Point in diff direction



$90^\circ \mid \cos(90^\circ) = 0$ ↳ Totally different dist

$\cos(45^\circ) = 1/\sqrt{2} = \frac{\sqrt{2}-1}{\sqrt{2}}$

Deep learning! XI/P

↓

Feature Engineering / feature selection

↓

Model. Main Key IDEA.

X I/P

↓

Deep learning

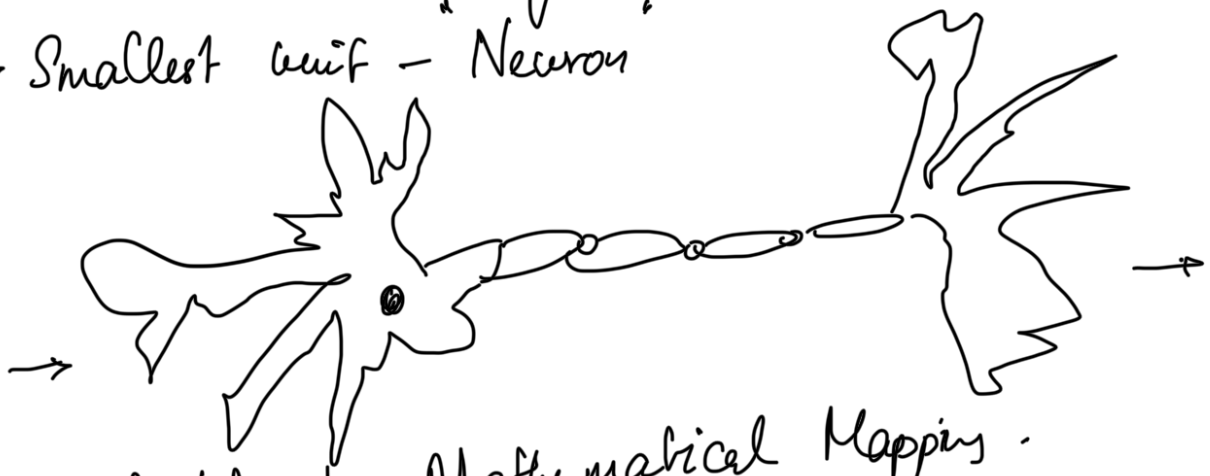
↓
Automatically Learn deep features
↓
classificati- / Regressi-

Deep learning uses Multi Layer "Neural Networks" to learn both useful representations & predictive relationships

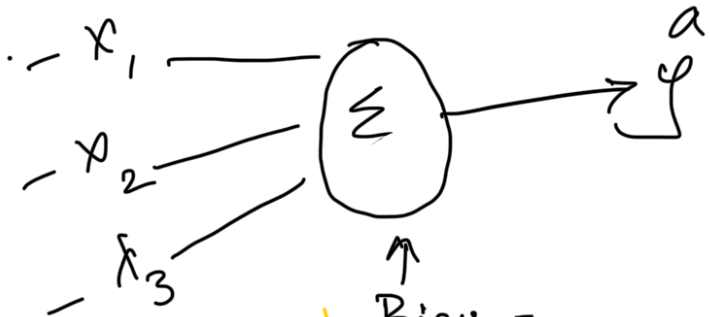
---

This contains Multi-Layered "Neural" Networks.

⇒ Smallest unit - Neuron



Neural Network Mathematical Mapping.



Weighted Sum

$$z = w_1x_1 + w_2x_2 + w_3x_3 + b$$

→ bias
Adjustable bias or threshold shift

Inputs
Weights

Input features
Columns



Salary

NOTE: I am not writing y

final Salary $z = w_1x_1 + w_2x_2 + 3.6\text{L} \overline{\text{E}}$

m.r.f. ✓

prev Sal ↓ exp. ↓

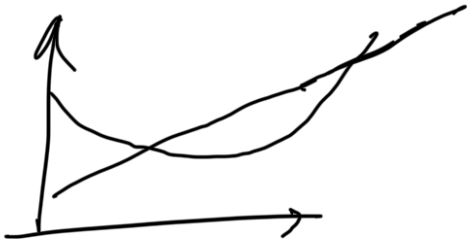
fresher ⇒ $h = w_1 x_1 + w_2 x_2 = 0$ +3.6 LPA

1 disadvantage here :

This System is Linear !!

w_1 | w_2 ⇒ Slopes of a St. line.

Polynomial Regression breaks linearity w.r.t I/p but it remains linear w.r.t models learned parameters.



Why is polynomial still considered a linear model?

$y = w_1 x + w_2 x^2 + b$

↓ ↓

x_1 x_2

$y = w_1 x_1 + w_2 x_2 + b$ ⇒ Proof of polynomial Still linear

---

Back to friendly neuron

$z = w_1 x_1 + w_2 x_2 + b$ →

⇒ $y = w_1 x_1 + w_2 x_1^2 + w \ldots$ - - -

x_1 | x_2 ⇒ $y = x_1 + x_2 + 2 x_1 x_2$

② Can ML models learn non-linear patterns?

Yes, they can via, Polynomial features, DTs, RFs, SVM → Kernels.

③ Why do we need Deep learning?

2 feats. | x_1, x_2

x_1, x_2 | $x_1^2\ x_2^2$ | $x_1 x_2$

$n \Rightarrow d$ 100

100 i/p

1000 Millions of pixels / words

$\binom{1000}{2} \Rightarrow 501,501$

Introduce & learn useful non-linear representations through trainable layers rather than manually specifying every pattern of feat

$$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

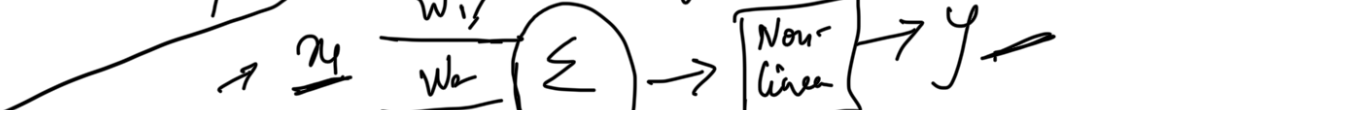
↓

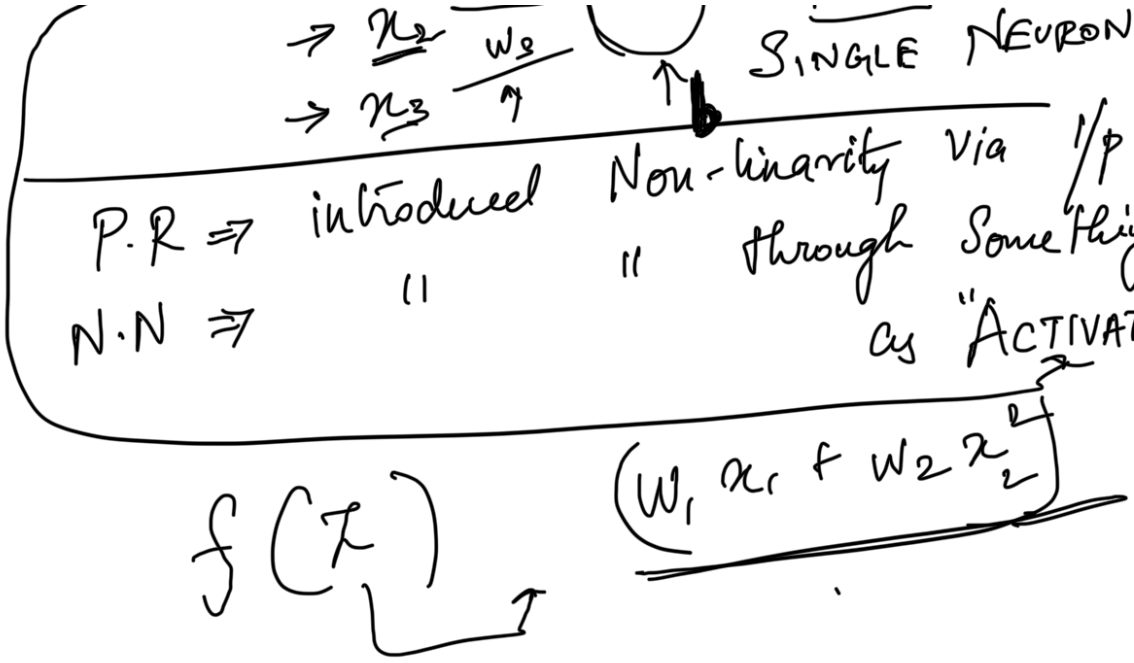
Non-linear function

↓

$\hat{y}$ of a Simple Single Neuron

$\hat{y} = f(z) \Rightarrow$





---

(4) When I have R.F which also does featu Imp. Why deep learning?

R.F is very useful if the features are Structured
↳ Tables.
When my data has bigga complexity like Language, Image or Speech

(1) R.F with mid featu → Time
(2) R.F won't look a previous Trees So learn complex patterns may be diff

---

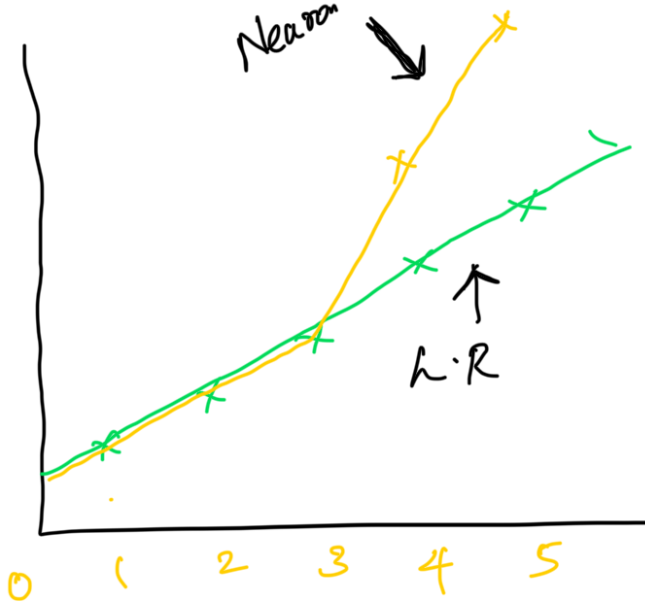
Company A 20k/months

Salary = 20,000 + 5,000 x

Exp	Salary
0	20000
1	25k

Exp	L.M	Actual.
0	20k	20k
1	25k	25k
2	30k	30k
3	35k	35k
4	40k	50k
5	45k	65k

2 3 30k 35k



ReLU

↳ Rectified linear Unit.

Starting Salary 5,000

20k ∫ ReLU = max (0, x)

$z = x - 3$

$1 \times x + (-3)$

$a = \text{ReLU}(z) = \max(0, x-3)$

Experience	Weighted Sum	ReLU Out.
1	-2	0
2	-1	0
3	0	0
4	1	1
5		

ReLU (x-3)

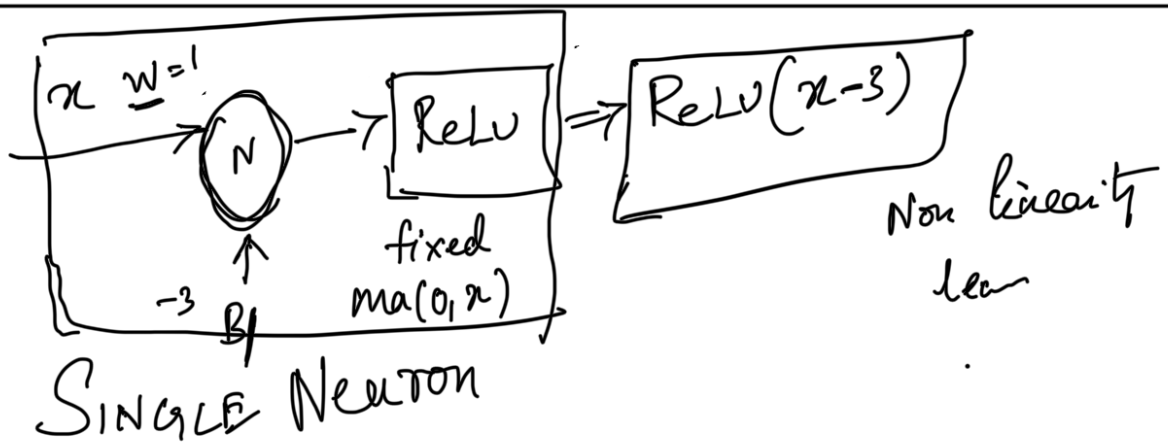
Yrs = 2 ⇒ Salary = 20,000 + 5,000x + 10,000 ReLU(x-3)

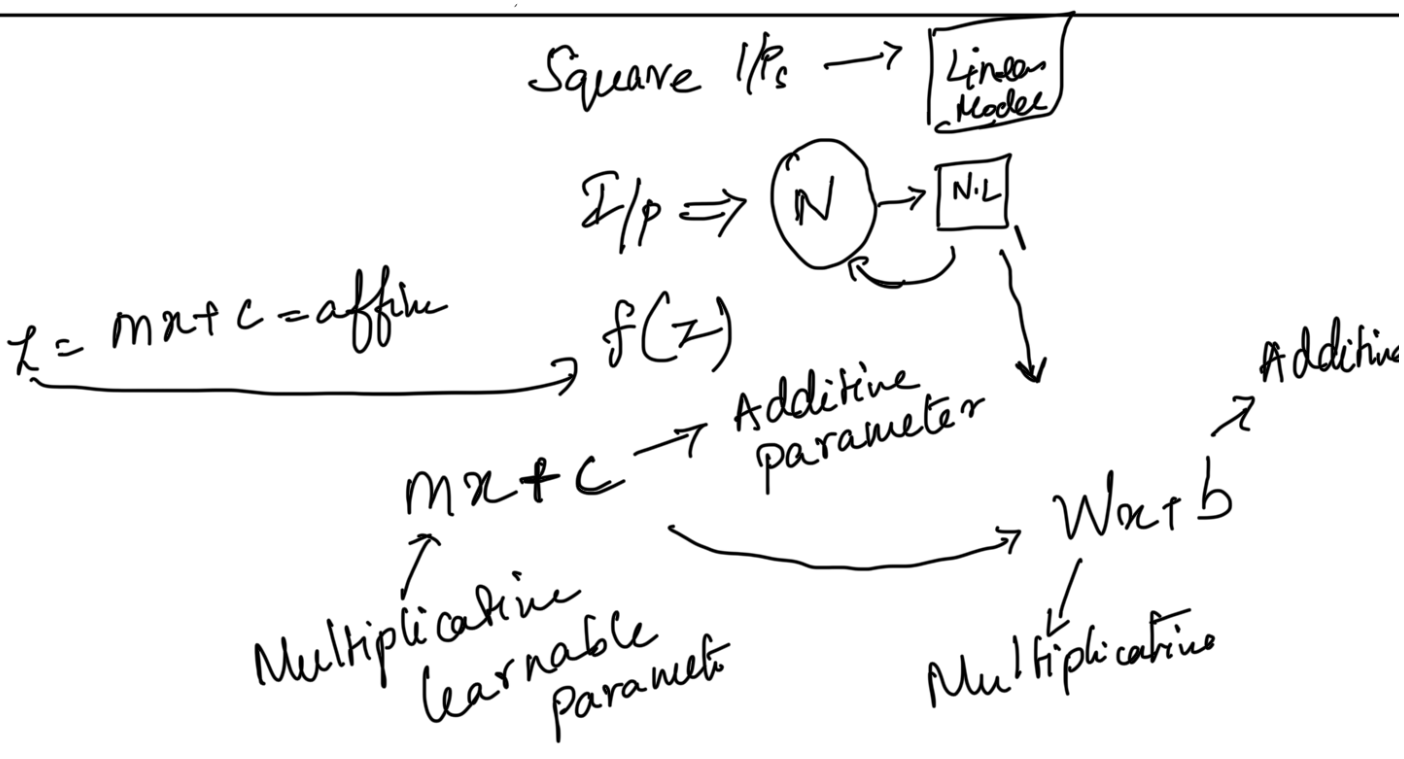
S = 30K

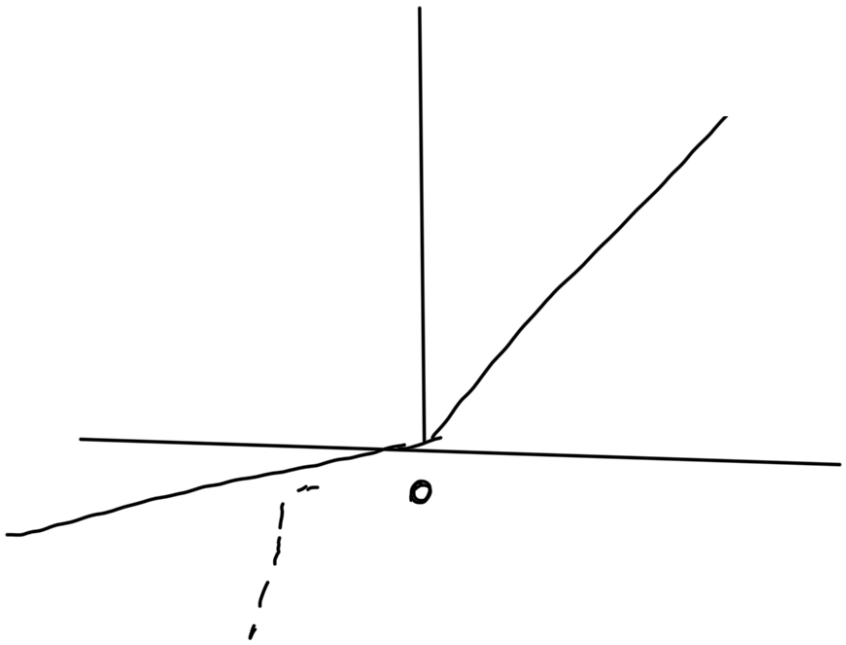
Yrs = 5 ⇒ y = 20,000 + 5000(5) + 10,000(2)

= 20,000 + 25000 + 20000

S = 65,000

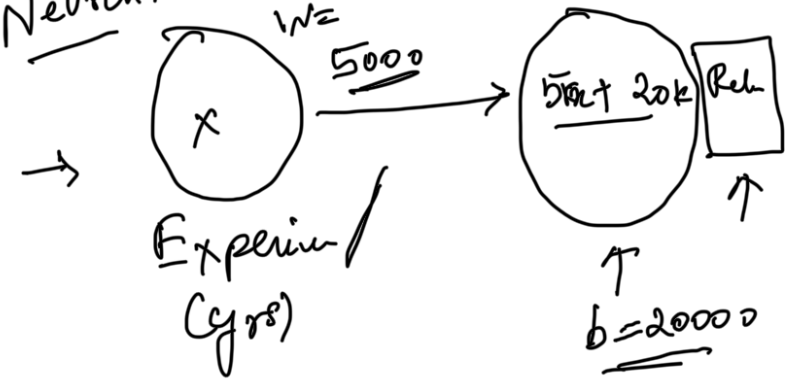






$S = 20{,}000 + 5000x + 10{,}000\ ReLU(x)$

Neuron



$ReLU(0, 5x+20) = 20{,}000$

$x = 0$

Neuron

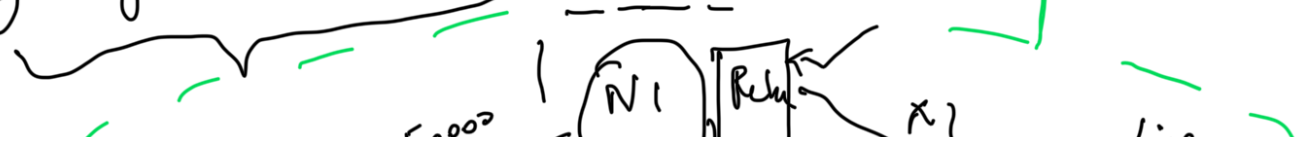


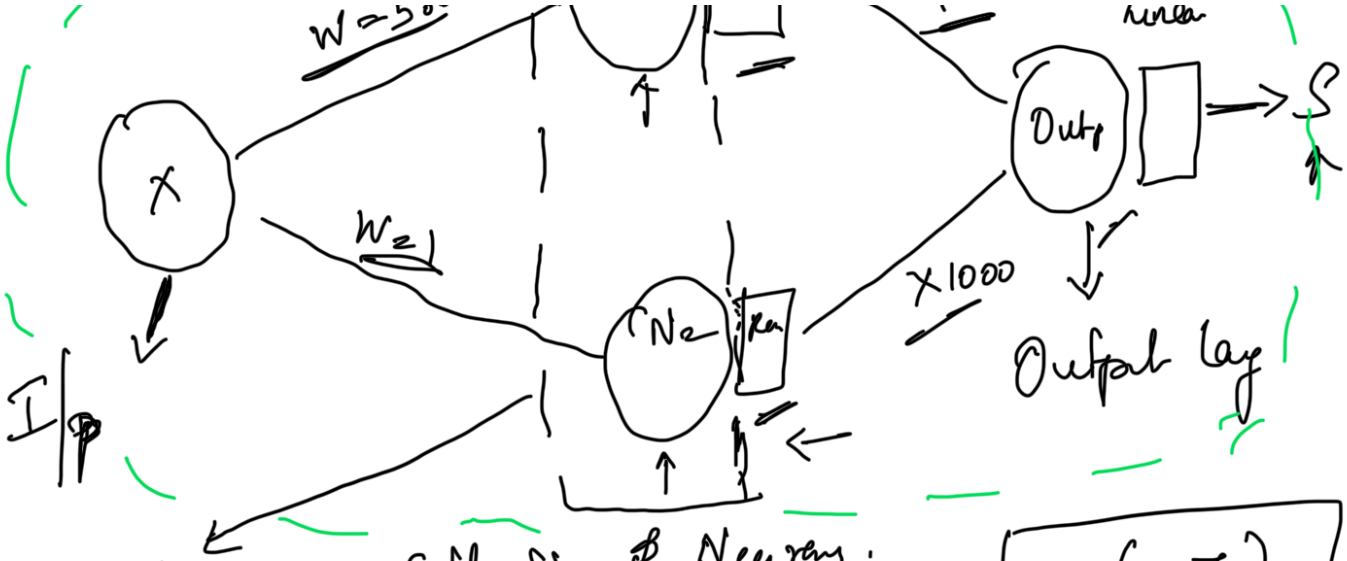
$ReLU(x)$

$ReLU(0, z)$

Terms

1. Neuron
2. Weights
3. biases
4. Activation fn.
5. Affine Trans $= mx+c$
6. Neural Network.





(7) Hidden Layer → Collection of Neurons.

$\max(0, z_i)$

Experience = 5

$h_1 = \text{Relu}(5(5) + 20) = 45$

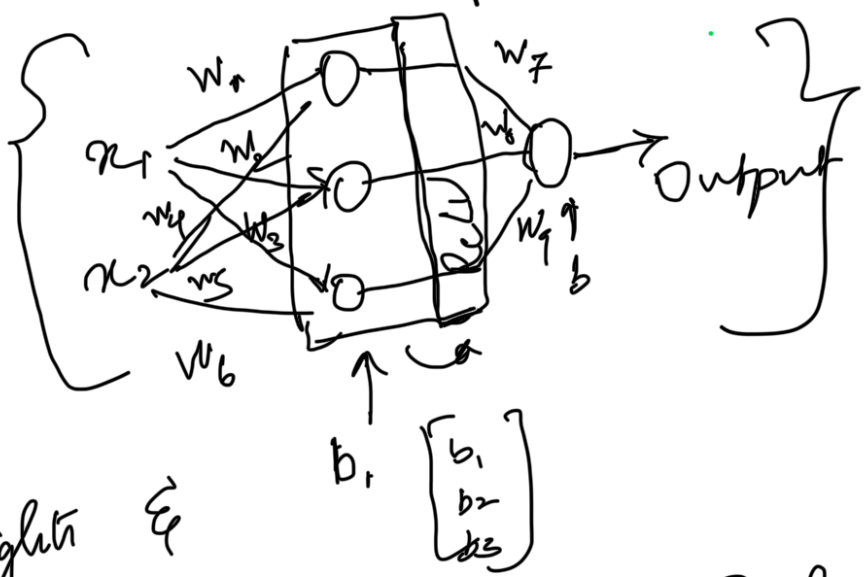
$h_2 = \text{Relu}(5 - 3) = 2$

---

final Sale:

$S = 1(h_1) + 10(h_2) + 0$

$= 45 + 10(2) = 65$



Global forms by additive & X^tive Mixing of I/p features

b_1 $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Weights & Bias are learnt via Optimization | Backprop.

(1)

Activation fn is the Non-linearity we add after each Hidden Layer.

Interview Questions

(1) What is the diffrnce bet ML & DL?
are DL Systems always better than ML?
If not what are i/p's where DL is better?

(2) Is Polynomial Linear?

Diff Poly & NN → Non-Model.

Non-linearity to i/p

(3) what introduces non-linearity in NNetworks? → Activation function